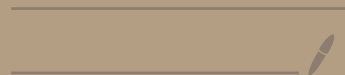


Labor-Demand Policies

Pascal Michailat

<https://pascalmichailat.org/v/>



Active labor market policies

Policies to reduce un-

-employment (when u is too high)

① Wage policies minimum wage, wage tax

② Public employment

Passive labor market policies

Policies to improve

situation of unemployed workers

③ Unemployment insurance

Minimum wage

Assume all workers are paid at

minimum wage W

What happens when minimum

wage goes up?

Use matching model w/ job rationing.

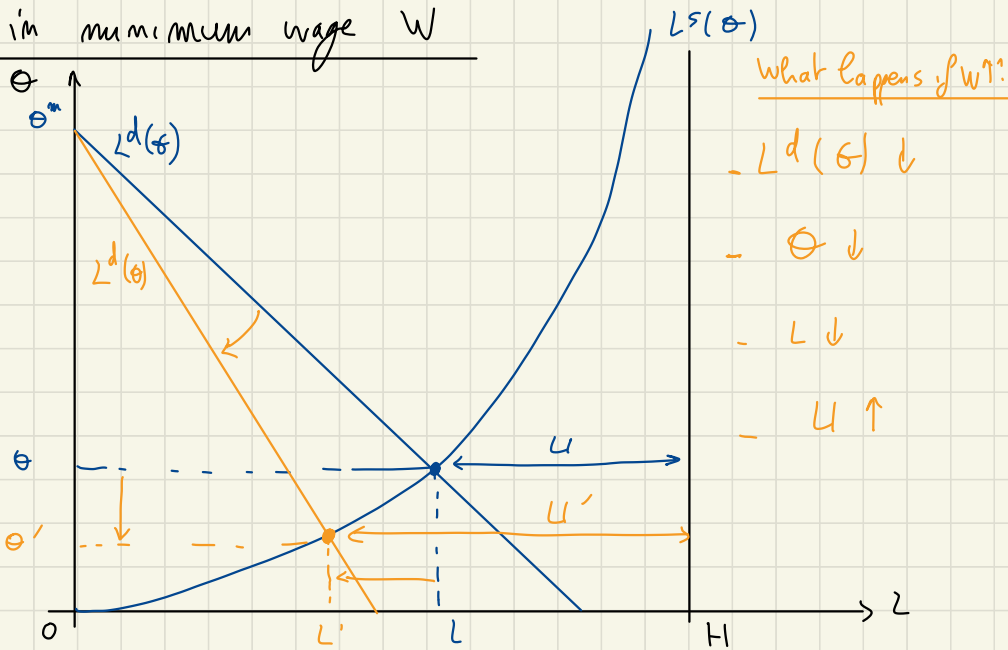
• Wage function W

• Production function. $Y = a N^\alpha$, $0 < \alpha < 1$

• Labor supply $L^S(\theta)$ unaffected by minimum wage

• Labor demand $L^D(\theta) = \left[\frac{\alpha a}{\underline{W} \cdot [1 + \tau(\theta)]^\alpha} \right]^{1/(1-\alpha)}$

Increase in minimum wage W



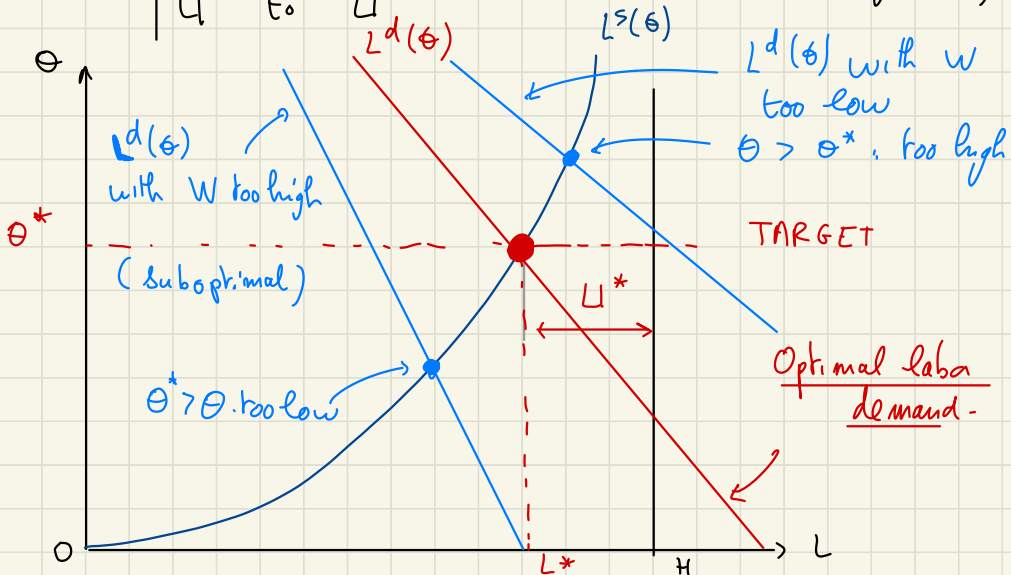
Optimal minimum wage

maximizes social welfare

Minimum wage that

brings θ to θ^* (efficient labor market tightness)

U to U^*



W^* optimal minimum wage $\rightarrow$ maximizes welfare
 • θ^* efficient tightness $\rightarrow \theta^*$ given by formula

• $\underline{L}^* = L^S(\theta^*) = \frac{f(\theta^*)}{1 + f(\theta^*)} \cdot H$

• $U^* = H - L^*$

• W^* is such that $L^d(W^*) = L^*$

$$\left[\frac{a \alpha}{W^* (1 + \tau(\theta^*))^\alpha} \right]^{1/(1-\alpha)} = L^*$$

$$\left[\frac{W^* (1 + \tau(\theta^*))^\alpha}{a \alpha} \right]^{1/(1-\alpha)} = \frac{1}{L^*}$$

$$\frac{W^* (1 + \tau(\theta^*))^\alpha}{a \alpha} = (L^*)^{\alpha-1}$$

$$W^* = \frac{a \alpha (L^*)^{\alpha-1}}{[1 + \tau(\theta^*)]^\alpha}$$

• If currently $\begin{cases} \theta < \theta^* \\ U > U^* \end{cases}$ then need reduce W to W^*

• If currently $\begin{cases} \theta > \theta^* \\ U < U^* \end{cases}$ then need increase W to W^*

Empirical evidence on minimum wage

Empirical literature is divided in 2 camps

Minimum wage reduces employment

Consistent with our model

Minimum wage has no effect on employment/unemployment.

Not consistent with our matching model $\rightarrow$ modify/improve the model to explain this fact - Need to introduce new elements such that minimum wage does not depress labor demand

① Efficiency - wage element: labor productivity increases w/ wage, $a = a(w)$ w/ $a'(w) > 0$

In labor demand $\frac{a}{w} = \frac{a(w)}{w}$

If $a(w)/w \sim \text{constant} \rightarrow w$ does not affect $L^d(\theta) \rightarrow$ minimum wage does not reduce employment -

△ can we still explain business cycles?

② Aggregate - demand elements $w \uparrow \Rightarrow$ disposable

income $\uparrow \Rightarrow$ spending $\uparrow \Rightarrow$ sales $\uparrow \Rightarrow$

"effective productivity" $\uparrow \Rightarrow a \uparrow$

Could introduce $a(w)$ w/ $a'(w) > 0$

Tax / subsidy or wage ($\sim$ change in payroll tax)

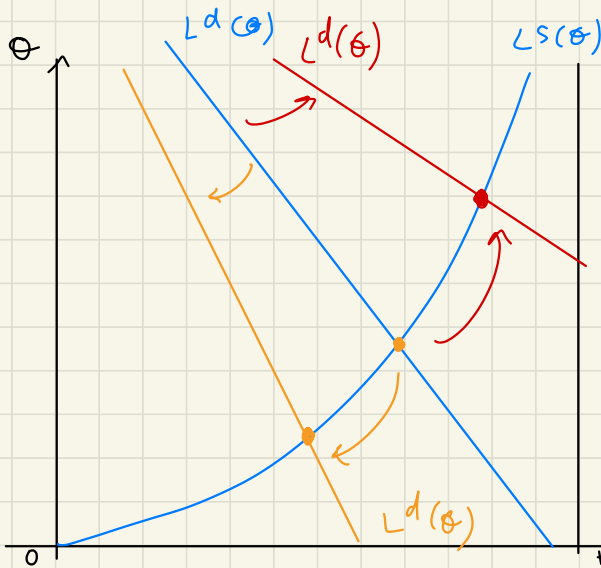
- W wage received by workers (wage net of tax)
- t payroll tax US. $t \approx 7\%$, funded UI system

Assume that firms pay payroll tax

- wage paid by firms $\cdot \underline{(1+t) \cdot W} > W$
- labor demand is modified by payroll tax

$$L^d(\theta) = \left[\frac{a \cdot \alpha}{\underline{(1+t)} W [1+\tau(\theta)]^\alpha} \right]^{1/(1-\alpha)}$$

- When $t \uparrow$ $L^d(\theta) \downarrow$ $\rightarrow$ same as increase in minimum wage
- When $t \downarrow$ $L^d(\theta) \uparrow$ $\rightarrow$ same as decrease in minimum wage



If $t \uparrow$

- $\theta \downarrow$
- $L \downarrow$
- $u \uparrow$

(same as $w \uparrow$)

If $t \downarrow$

$\rightarrow L$

Optimal payroll tax t^* (to maximize welfare
→ reach θ^*)

efficiency θ^* , $L^* = L^S(\theta^*)$, $U^* = H - L^*$

Optimal payroll tax such that $L^d(\theta^*) = L^S(\theta^*) = L^*$

Solve $L^d(\theta^*) = L^*$

$$\Rightarrow \left[\frac{a \alpha}{(1+t^*) w (1+\tau(\theta^*))^\alpha} \right]^{1/(1-\alpha)} = L^*$$

$$\Rightarrow \frac{(1+t^*) w [1+\tau(\theta^*)]^\alpha}{a \alpha} = (L^*)^{\alpha-1}$$

$$\Rightarrow 1+t^* = \frac{a \alpha (L^*)^{\alpha-1}}{[1+\tau(\theta^*)]^\alpha w}$$

t^* could be > 0 or < 0 .

Δ If payroll tax paid by firms (incidence of tax is on firms): payroll is effective tool

But if payroll tax paid by workers (incidence of tax is on workers): firms & labor demand are unaffected by tax → tax is completely ineffective.

Public employment

- US
- # workers in public sector = 17% of # workers in economy
 - spending on public workers = 63% of government spending
 - stimulus packages often raise public employment Example US New Deal

Introducing public employment in matching model

- Matching process. public & private workers are part of same labor market
 - V : # vacancies from firms + # vacancies from government
 - $\theta = V/U$
 - s job-separation rate applies both to private firms & government
 - $m(U, V)$ gives # matches in aggregate labor market (firms + government)
 - workers apply indiscriminately to public & private jobs
 - government & private firms recruit workers indiscriminately

- Labour supply not affected by public employment.

$$L^S(\theta) = \frac{f(\theta)}{\Delta + f(\theta)} H$$

private employment + public employment | aggregate employment

Labour demand is modified by public employment.

Aggregate labour demand

= Private labour demand + public labour demand
(by firms) (by government)

= $L^d(\theta)$ + G

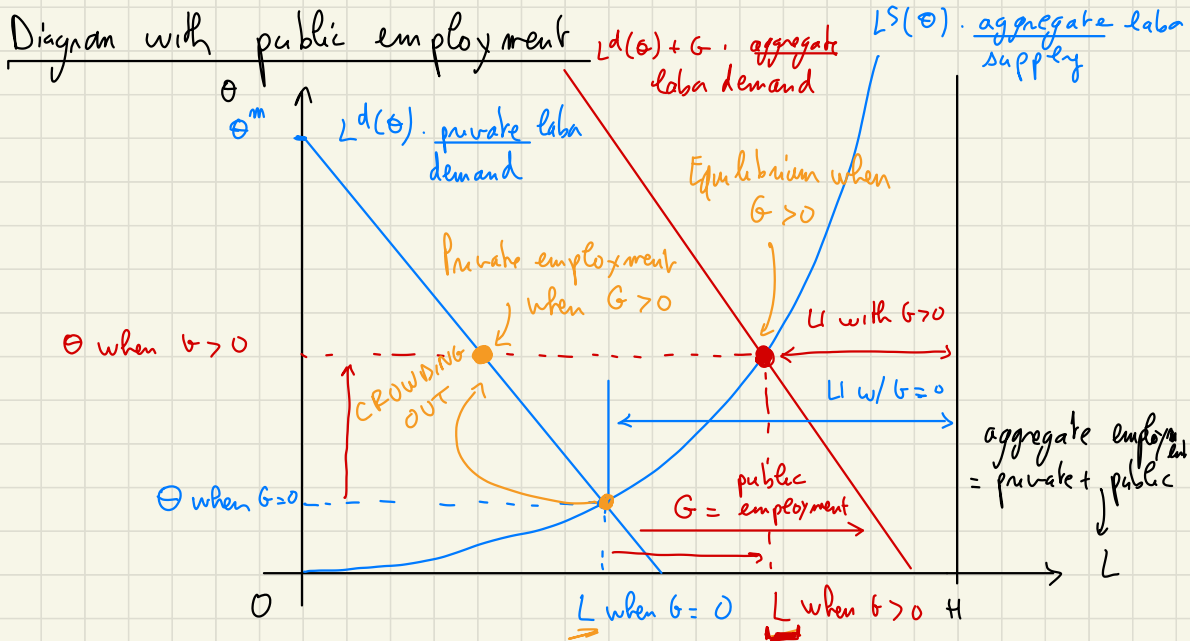
$$= \left[\frac{\alpha \cdot a^{1-\alpha}}{w [1 + \tau(\theta)]^\alpha} \right]^{1/(1-\alpha)} + G$$

• Production function is concave,
 $y = a \cdot N^\alpha$ $0 < \alpha < 1$

• Wage is rigid
 $w = \bar{w} a^\delta$ $0 \leq \delta < 1$

Labour market equilibrium

$$\underbrace{L^d(\theta) + G}_{\text{aggregate labour demand}} = \underbrace{L^S(\theta)}_{\text{aggregate labour supply}}$$



Introduction of public employment $G > 0$ leads to

- Aggregate employment $L \uparrow$
- Tightness $\Theta \uparrow$
- Unemployment $U \downarrow$
- Private employment $L^d(\Theta) \downarrow$

→ there is crowding out of private employment by public employment

Public - employment multiplier λ

Definition additional # of workers employed when 1 worker is hired in public sector

$$\lambda = \frac{dL}{dG}$$

Computation of λ

$$\begin{array}{l} G=0 \quad L^d(\theta) = L^s(\theta) \\ G>0 \quad L^d(\theta) + \underline{G} = L^s(\underline{\theta}) \end{array}$$

Implicitly, θ is a function of G through equilibrium condition.

Increase public employment by $dG > 0$

- Employment changes by dL
- Tightness changes by $d\theta$

① Compute $d\theta$

② Infer $dL \rightarrow \lambda = dL/dG$

$$\underbrace{L^d(\theta) + G}_{\text{LHS}} = \underbrace{L^s(\theta)}_{\text{RHS}}$$

$dG \rightarrow d\text{LHS} \text{ \& \& } d\text{RHS}$

Since equilibrium condition is valid before & after change dG , then $d\text{LHS} = d\text{RHS}$

$$\bullet \text{dRHS} = \frac{dL^S}{d\theta} \cdot d\theta$$

$$\bullet \text{dLHS} = \frac{dL^d}{d\theta} \cdot d\theta + dG$$

$$\text{Hence } \frac{dL^d}{d\theta} d\theta + dG = \frac{dL^S}{d\theta} d\theta$$

$$\left[\frac{dL^S}{d\theta} - \frac{dL^d}{d\theta} \right] d\theta = dG$$

$$\left[\frac{dL^S}{d\theta} - \frac{dL^d}{d\theta} \right] \frac{d\theta}{dG} = 1$$

$$\left| \frac{d\theta}{dG} = \frac{1}{\frac{dL^S}{d\theta} - \frac{dL^d}{d\theta}} \right.$$

Recall from "Unemployment fluctuations"

$$\bullet \varepsilon_{\theta}^{L^S} = (1-\eta) \cdot u(\theta) \quad |$$

$$\varepsilon_{\theta}^{L^S} = \frac{d \ln L^S}{d \ln \theta} = \frac{\theta}{L^S} \frac{dL^S}{d\theta}$$

$$\Rightarrow \frac{dL^S}{d\theta} = \frac{L^S}{\theta} \cdot \varepsilon_{\theta}^{L^S} = \frac{L}{\theta} \cdot \varepsilon_{\theta}^{L^S}$$

$$\boxed{\frac{dL^S}{d\theta} = \frac{L}{\theta} (1-\eta) u}$$

Assumption Cobb-Douglas matching function

$$\cdot \varepsilon_{\theta}^{L^d} = -\frac{\alpha}{1-\alpha} \eta \tau$$

$$\varepsilon_{\theta}^{L^d} = \frac{d \ln L^d}{d \ln \theta} = \frac{\theta}{L^d} \frac{d L^d}{d \theta}$$

$$\Rightarrow \frac{d L^d}{d \theta} = \frac{L^d}{\theta} \cdot \varepsilon_{\theta}^{L^d} = \frac{L-G}{\theta} \cdot \varepsilon_{\theta}^{L^d}$$

$$\Rightarrow \frac{d L^d}{d \theta} = -\frac{\alpha}{1-\alpha} \cdot \eta \tau \frac{(L-G)}{\theta}$$

Thus

$$\frac{d \theta}{d G} = \frac{1}{\frac{L}{\theta} (1-\eta) u + \frac{\alpha}{1-\alpha} \eta \tau \frac{(L-G)}{\theta}} > 0$$

Multiply, $\lambda = \frac{d L}{d G}$ and $L = L^S(\theta)$

$$\lambda = \frac{d L}{d G} = \frac{d L^S}{d \theta} \cdot \frac{d \theta}{d G}$$

$$\lambda = \frac{(L/\theta) (1-\eta) u}{(L/\theta) (1-\eta) u + \frac{\alpha}{1-\alpha} \cdot \eta \cdot \tau \left(\frac{L-G}{\theta} \right)}$$

$$\lambda = \frac{1}{1 + \frac{\alpha}{1-\alpha} \frac{\eta}{1-\eta} \frac{\tau}{u} \left(\frac{1-G}{L} \right)}$$

share of workers
in private
sector σ

shape of
production function

shape of
matching
function

$$\lambda = \frac{dL}{d\sigma} = \frac{1}{1 + \frac{\alpha}{1-\alpha} \frac{n}{1-\eta} \frac{\tau(\theta)}{u(\theta)} \sigma}$$

Properties of the public-employment multiplier

① $\lambda > 0$.

When public employment $\uparrow$, then $\left\{ \begin{array}{l} \text{total employment } \uparrow \\ \text{unemployment } \downarrow \end{array} \right.$

② $\lambda < 1$.

Total employment increases less than public employment - Unemployment decreases by less than public employment increases -

This is b/c private employment is crowded out.

When public employment $\uparrow$, private employment $\downarrow$,
so total employment $\uparrow$ by less than public employment.

→ crowding out b/c $\theta \uparrow$ when $\sigma \uparrow$.

③ λ is countercyclical. λ is large when θ

(L) is low but λ is low when θ is high.

Good times, low λ / Bad times (L) high λ .

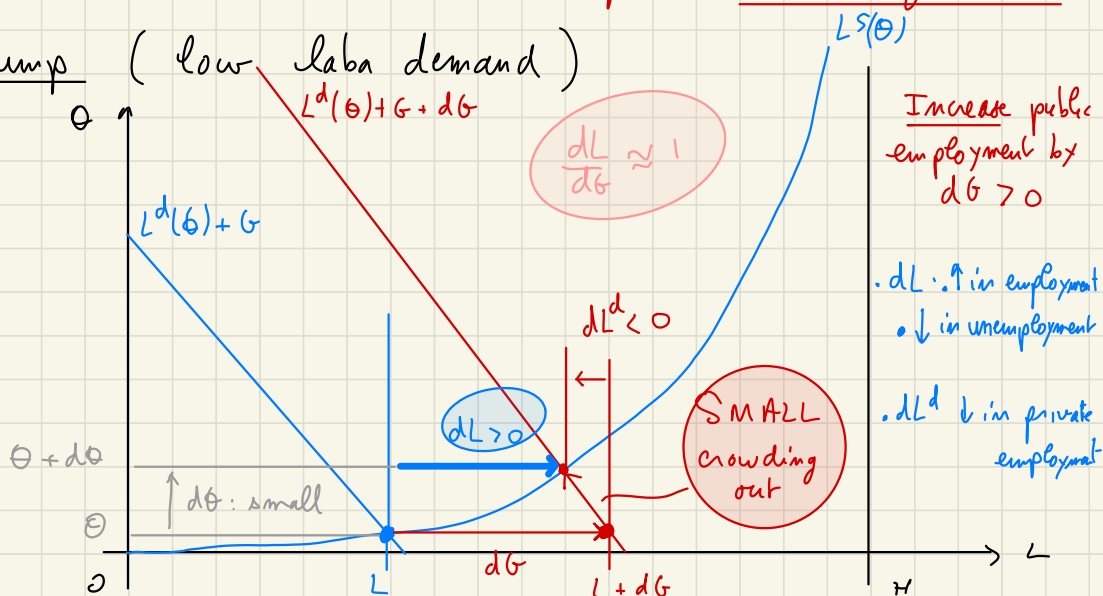
$$\lambda(\theta) = \frac{1}{1 + \frac{\lambda}{1-\lambda} \frac{\eta}{1-\eta} - \sigma} \frac{\tau(\theta)}{u(\theta)}$$

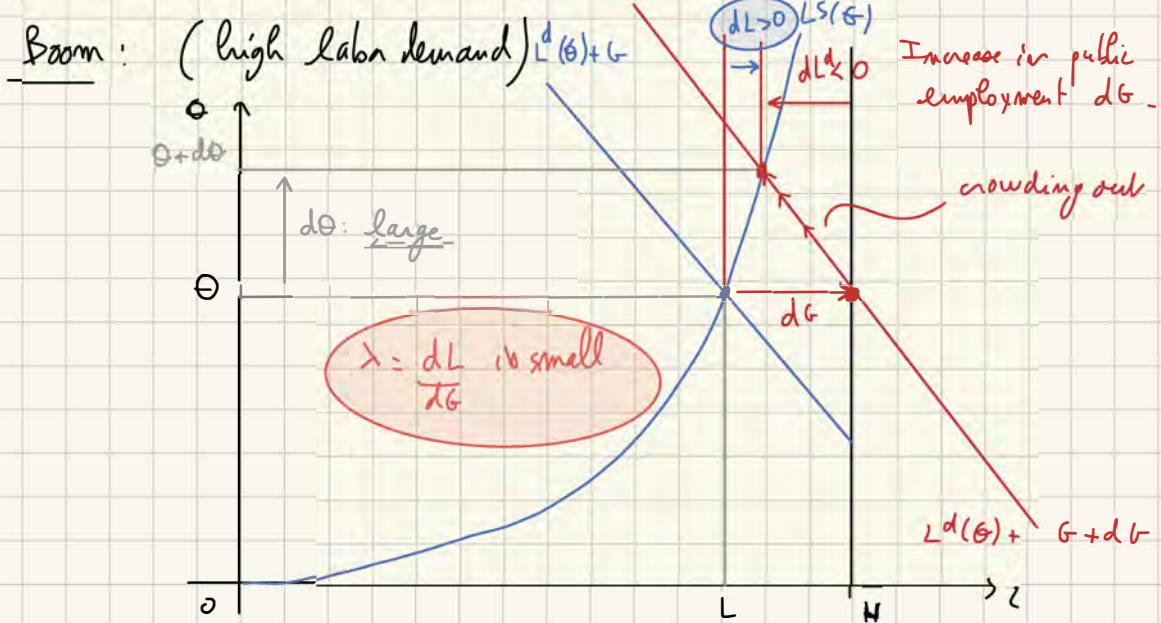
good times } $\frac{\tau(\theta)}{u(\theta)}$ is high $\Rightarrow \lambda(\theta)$ is low
(θ is high) } $\frac{\tau(\theta)}{u(\theta)}$ is low

bad times
 $\frac{u(\theta)}{\tau(\theta)}$ } $\frac{\tau(\theta)}{u(\theta)}$ is low $\Rightarrow$ $\lambda(\theta)$ is high
 (θ is low)

$\lambda'(\theta) < 0$ so multiplier is countercyclical

Stamps (low labor demand)





2